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Project Title:	Assessing the Impact of Caffeine and Other Dietary Factors on Crew Performance and Sleep		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
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Organization Name:	NASA Ames Research Center		
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PI Address 2:	Human Systems Integration Division, Code 262-4		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
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Key Personnel Changes/Previous PI:	n/a		
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Task Description:	Statement of the Problem: Humans require 7-8 hours of sleep for adequate cognitive function and behavioral health. Chronic sleep deprivation is associated with progressive performance impairment with each day of insufficient sleep. Astronaut crews have historically averaged around 6 hours of sleep per night, especially during missions that required a high tempo work environment, confined spacecraft without crew quarters, and frequent sleep schedule changes. Each of these situations are likely to occur during Artemis and Mars missions. While such short sleep duration has been shown to reduce crew alertness and performance on the International Space Station (ISS), crewmembers typically perform better than average individuals when they are sleep deprived. It is possible that this better than average performance is due to resilience against the negative impacts of sleep loss. However, it is also possible that astronaut crews self-select countermeasures, such as caffeine, that compensate for the performance decrements that typically accompany sleep loss. Caffeine is the most widely used performance-enhancing drug on Earth. Astronaut crews have access to caffeine in the form of pills and liquid coffee. Numerous studies have confirmed the utility of caffeine as a countermeasure against the effects of sleepiness and neurobehavioral performance. For example, chronic low dose caffeine consumption (0.3 mg/kg/h) can sustain performance over 28 hours of sleep deprivation. While caffeine is clearly a potent countermeasure to improve alertness and performance on a variety of tasks, it also interferes with sleep. This leads to performance deficits on the following day, driving a cycle of caffeine use to counter the effects of caffeine-induced sleep disruption. There are few data available on the use of caffeine by astronauts during spaceflight. In post-flight interviews, 75% of astronauts reported use of alertness medications during their mission, including caffeine pills or modafinil and ISS crewmembers reported using caffeine on more than 90% of days before and during long duration missions, regardless of whether or not they were circadian misaligned. Habitual, rather than strategic, caffeine use and the presumably near-constant presence of such a powerful performance enhancing drug likely masks the true extent of sleepiness due to chronic sleep loss and circadian misalignment during spaceflight and analog missions. We therefore have little information on what would happen to sleepiness and performance in operational settings if caffeine became unavailable during a long-duration mission, and plan for that contingency. Current evidence suggests that the impact would be quite profound. Caffeine availability will be limited during future Artemis and Mars missions. For example, Orion will not have hot water, making liquid coffee unavailable as a countermeasure. Similarly, longer duration missions will be unlikely to have sufficient payload to provide crews continuous access to caffeine. As a result of these mission constraints, it is critical that we 1) determine the prevalence of caffeine use among crews, 2) characterize the impact of caffeine on performance, and 3) characterize the impact of caffeine on sleep. This information will provide us with an understanding of how caffeine is being used inflight to help guide countermeasure use in future missions. Specific Aims: 1) We will characterize the use of caffeine (frequency, timing, dose, proportion of caffeine users) among 21 crewmembers during ISS missions in order to determine how much caffeine would be required to maintain current use; 2) We will evaluate the impact of caffeine use on cognitive performance by comparing performance following caffeine ingestion to performance on days when no caffeine was consumed. We will make these comparisons both between and within individuals to determine how caffeine impacts performance. We will further evaluate whether cognitive dimensions are affected differentially by caffeine (e.g., better reaction time, but with increased impulsivity following caffeine use); 3) We will characterize the impact of caffeine use on sleep duration, latency, and wake after sleep onset on days when caffeine is consumed compared to days when no caffeine was consumed. We will further evaluate the impact of later timing of caffeine consumption on sleep outcomes; and 4) We will conduct exploratory analyses to evaluate links between other dietary factors on sleep, including intake of macronutrients, such as carbohydrates. Research Methods: No new data will be collected as part of this task. We will combine actigraphy, survey, and Cognition test battery data from the Standard Measures protocol with caffeine data from the Medical Evaluation Documents (MED) Volume B (MedB) ISS Food Intake Tracker (FIT) application.
Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation. The results of this directed task analysis are needed quickly to inform early Artemis missions.
Research Impact/Earth Benefits:	This information will add to our understanding about how caffeine affects sleep and performance in humans. This information is important for determining spaceflight payloads and for understanding how crew members may be self-selecting fatigue countermeasures.
Task Progress:	The data were requested and approved through the Life Sciences Data Archive (LSDA) board and the NASA Institutional Review Board (IRB). Mean daily caffeine and nutrition data were received in July of 2023. These data were analyzed to identify the number of participants who consumed caffeine, the average amount of caffeine consumed each day, and the relationship between caffeine and prior and subsequent sleep. These findings were presented at the 2024 NASA Human Research Program Investigators' Workshop (HRP IWS) meeting. No performance or detailed caffeine information was provided as part of the initial analysis; therefore, an extension was requested for the project. Detailed information about the type of caffeine consumed each day was provided in June 2024. Upon comparing the detailed file with the average file that was originally provided, we determined that the original file was not accurate. After investigation of this issue with the nutritional biochemistry laboratory, we identified that caffeine amounts were not correctly estimated (i.e., regular coffee had a lower caffeine value listed than decaf coffee). As a result, a second extension was requested to review all of the data that was provided to us and to complete new analyses for a period of performance ending in June 2025. We found that caffeine use is prevalent during International Space Station (ISS) missions, with all crew members assessed consuming caffeine during their mission. This resulted in a negative impact on sleep, with increasing caffeine consumption on a given day leading to a small but significant increase in wake time during sleep. Correspondingly, disrupted sleep resulted in more caffeine use the following day. These findings confirm that caffeine use contributes to a cycle of subsequent sleep loss and more caffeine use. We also found that caffeine use improved processing speed for cognitive tests requiring executive function, but not simple reaction time or accuracy. This suggests that caffeine reduces

	time on task among high performers and is likely helping to compensate for some aspects of sleep loss during spaceflight.
Bibliography Type:	Description: (Last Updated: 08/01/2025)
Articles in Peer-reviewed Journals	Flynn-Evans EE, Braun AM, Jansen RA. "Sleep away from Earth." Sleep Med Clin. 2025 Mar;20(1):73-80. https://doi.org/10.1016/j.jsmc.2024.10.003 ; PubMed PMID: 39894600 , Mar-2025